Workshop: find and fix the reasoning

LEARNING TARGET I can use a picture to identify a mistake and explain a correction.

A · KEY VOCABULARY

Distance vs. displacement	Trace the whole path vs. join start to finish.
Net force first	Combine all forces before dividing by mass.
Sign = direction	A minus sign is not "slow." Compare sizes for speed.
Pair forces	Act on different objects, so they never cancel.

B · KEY ROUTINE

1. Identify the first mistake
2. Repair it with a drawing + rule
3. Recheck on a new problem

evidence point to a graph or diagram
reason name the physics rule

C · COLOR IT · FOUR COMMON MISTAKES

Lightly shade each strip.

1. **CLAIM ✗** "3 m right then 3 m back — the distance is zero."
2. **FIX ✓** Distance = 3 + 3 = 6 m. *Displacement* is 0 m: the endpoints match.

1. **CLAIM ✗** "8 N pull on a 2 kg cart: 8 ÷ 2 = 4 m/s²."
2. **FIX ✓** Net force first: 8 − 2 = 6 N. Then 6 ÷ 2 = 3 m/s².

1. **CLAIM ✗** "−2 → −6 m/s is slowing, because it's negative."
2. **FIX ✓** Speed grows from 2 to 6 m/s: speeding up. The minus sign is direction.

1. **CLAIM ✗** "The force pair cancels, so nothing moves."
2. **FIX ✓** Each pair force acts on a different cart. Draw one on each diagram.

☐ **1 · PENCIL**
Mistaken claims: cross out

☐ **2 · CODE COLOR**
The fixes: color by quantity

D · WORKED EXAMPLE

A 3 kg cart has 9 N right and 3 N left. A student writes 9 ÷ 3 = 3 m/s². Correct it.

GIVEN	EQUATION	WORK	ANSWER
m = 3 kg 9 N right 3 N left	$a = F_{net} \div m$	$F_{net} = 9 - 3 = 6$ N right $a = 6 \div 3$	**2 m/s² right** missed: net force first

E · THE 2026-XJ THREAD

DAY 19 OF 22 · NEW TOOL: CHECKING REASONING

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else